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ORIGINAL RESEARCH ARTICLE

Thermodynamic Description of the Gadolinium-Cadmium Phase Diagram

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Abstract The thermodynamic optimization of the Gd-Cd binary system was carried out with the help of CALPHAD (CALculation of PHAse Diagram) method., $GdCd_3$, $Gd_{13}Cd_{58}$ and $GdCd_8$ have been treated as stoichiometric compounds while a solution model has been used for the description of the liquid, HCP_A3 (αGd), BCC_A2 (βGd) and HCP_A3 (Cd) phases. The intermetallic compounds GdCd,α-$GdCd_2$, β-$GdCd_2$, $Gd_{11}Cd_{45}$ and $GdCd_6$, which have homogeneity ranges, were treated by a two-sublattice model. The calculations based on the thermodynamic modeling are in a good agreement with the phase diagram data and experimental thermodynamic values.

Keywords CALPHAD method · Gd-Cd system · phase diagram · Redlich–Kister equation

1 Introduction

The alloy systems or intermetallic compounds composed of cadmium have attracted the attention for the development of lead-free soldering materials for new electronic appliances. Cadmium-based alloys are currently regarded as promising candidates for use as negative electrode materials in batteries.[1]

In order to define the processing conditions for making these alloys and subsequent treatments to obtain the optimum engineering properties, a knowledge of the phase diagram and thermodynamic properties of these alloys is essential. This present work deals with a reassessment of the thermodynamic description of the Gd-Cd system using the Calphad method.[2] In this method, the thermodynamic models for the Gibbs energy of all the individual phases are studied using the Parrot module in the Thermo-Calc package.[3] The thermodynamic parameters involved in the models are optimized using the experimental thermodynamic and phase diagram data.

2 Review of Experimental Data

The Gd-Cd equilibrium phase diagram was firstly investigated by Bruzzone et al.[4] using differential thermal analysis, x-ray diffraction, chemical and metallographic analyses. The corresponding phase diagram is shown in Fig. 1. Phase equilibria and melting behavior were determined in the complete composition range and altogether six intermetallic compounds were described: GdCd, $GdCd_2$, $GdCd_3$, $GdCd_{4\sim}$, $GdCd_{4.5\sim}$ and $GdCd_6$. No detectable range of solid solubility was observed for any of the phases except for the high-temperature allotropic modification of Gd. A polymorphic transformation (α ↔ β) was indicated at 995 °C for $GdCd_2$.

Gschneidner and Calderwood[5] proposed and assessed of the Gd-Cd phase diagram based on the work of Bruzzone et al.[4] These authors described the compounds $GdCd_{4\sim}$ and $GdCd_{4.5\sim}$ with the consistent notation

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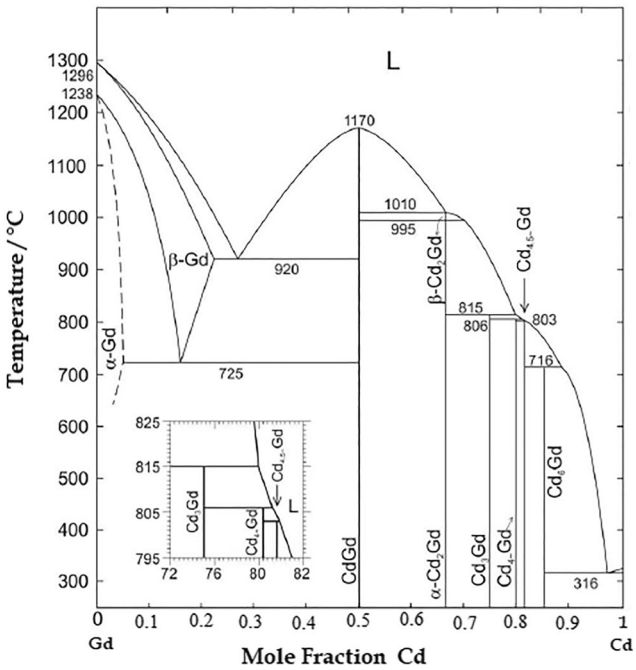

Fig. 1 Experimental Gd-Cd phase diagram proposed by Bruzzone et al.[4] Reprinted from *Journal of the Less Common Metals,* Vol 617, G. Bruzzone, M.L. Fornasini, F. Merlo, The gadolinium-cadmium system, Copyright 1971, with permission from Elsevier

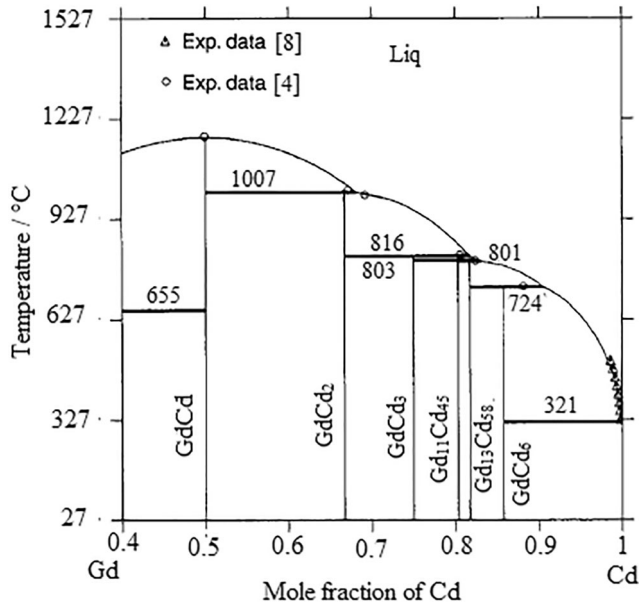

Fig. 2 Calculated Gd-Cd phase diagram according to Kurata and Sakamura[7] and compared with the experimental data (○ Bruzzone et al.,[4] Δ Johnson[8]). Reproduced with permission from Masaki Kurata et al., Thermodynamic assessment of systems of actinide or rare earth with Cd, *Journal of Phase Equilibria,* volume 22, pages 232–240, 2001, Springer Nature

$Gd_{11}Cd_{45}$ and $Gd_{13}Cd_{58}$, which is also commonly used in other Cd-RE systems. The latter phase diagram was accepted by Massalski et al.[6] Six intermediate phases existing in this system were shown as stoichiometric compounds, GdCd, $GdCd_2$, $GdCd_3$, $Gd_{11}Cd_{45}$, $Gd_{13}Cd_{58}$ and $GdCd_6$.

Later, Kurata and Sakamura[7] attempted thermodynamic modeling of the Gd-Cd binary systemwith the Thermo-Calc software[3] in the composition range 0 to 60 at.% Gd. All intermetallic compounds were modeled as stoichiometric compounds and compared with the experimental data by Bruzzone et al.[4] and Johnson.[8] The results are reproduced in Fig. 2.

Recently, the Gd-Cd phase diagram was reinvestigated by Reichmann and Ipser,[9] using x-ray powder diffraction (powder-XRD), scanning electron microscopy (SEM), and differential thermal analysis (DTA). The result is shown in Fig. 3. A new compound was reported a stoichiometry of "$GdCd_8$", decomposing peritectically at 465 °C. No crystal structure data were available for this phase. These authors confirmed the existence of six intermediate phases: GdCd, $GdCd_2$ (a polymorphic transformation $\alpha \leftrightarrow \beta$ at 986 °C), $GdCd_3$, $Gd_{11}Cd_{45}$, $Gd_{13}Cd_{58}$ and $GdCd_6$ which were already described by Bruzzone et al.[4] Four intermetallic compounds: GdCd, $GdCd_2$, $Gd_{11}Cd_{45}$ and $GdCd_6$ are characterized by homogeneity ranges. In addition, the peritectic decomposition of $Cd_{58}Gd_{13}$ could not be

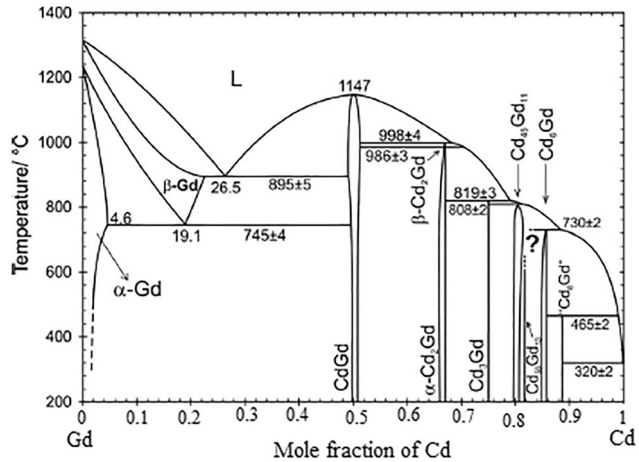

Fig. 3 The phase diagram of the Gd-Cd system reviewed by Reichmann and Ipser.[9] Reproduced from *Journal of Alloys and Compounds* under the terms of the Creative Commons Attribution license

confirmed as given by Bruzzone et al.[4] The crystal structure data of this system are presented in Table 1.

The standard enthalpies of formation and Gibbs energies of formation of the phase $GdCd_6$ were measured by Roshchina and Bayanov[10] using the electromotive force (E.M.F) method. Later, Kurata and Sakamura[7] calculated the enthalpies of formation for six intermetallic compounds: GdCd, $GdCd_2$, $GdCd_3$, $Gd_{11}Cd_{45}$, $Gd_{13}Cd_{58}$ and $GdCd_6$. by the Calphad method.[2] Recently Reichmann

Table 1 Crystal structures of the stable solid phases in the Gd-Cd system

Phase	Composition at.% Cd	Pearson symbol	Space group	Symbol used in Thermo-Calc data file	Proto-type	Reference
(α-Gd)	0 to ~ 5	*hP*2	$P6_3/mmc$	*HCP_A3*	Mg	[4]
(β-Gd)	0 to ~ 22	*cI*2	$Im\bar{3}m$	*BCC_A2*	W	[4]
CdGd	~ 49.8 to ~ 50.9	*cP*2	$Pm\bar{3}m$	GDCD	CsCl	[4]
β- Cd_2Gd	66.7	…	…	$GDCD_2$_HT	…	[4]
α- Cd_2Gd	65.3 to 67.1	*hP*3	$P6/mmm$	$GDCD_2$_BT	AlB_2	[9]
Cd_3Gd	75	*hP*8	$P6_3/mmc$	$GDCD_3$	Ni_3Sn	[4]
$Cd_{45}Gd_{11}$	~ 79.6 to ~ 80.8	*cF*448	$F\bar{4}3m$	$GD_{11}CD_{45}$	$Cd_{45}Sm_{11}$	[4]
$Cd_{58}Gd_{13}$	81.7	*hP*142	$P6_3/mmc$	$GD_{13}CD_{58}$	$Pu_{13}Zn_{58}$	[4]
Cd_6Gd	~ 84.8 to ~ 85.7	*cI*168	$Im\bar{3}$	$GDCD_6$	Cd_6Y	[4]
Cd_8Gd	88.7	…	…	$GDCD_8$	…	[9]
(Cd)	100	*hP*2	$P6_3/mmc$	*HCP_A3*	Mg	[4]

et al.[11] calculated the standard Gibbs energies of formation of the phases $GdCd_6$, $Cd_{58}Gd_{13}$, $Cd_{45}Gd_{11}$, Cd_3Gd and Cd_2Gd by means of Gibbs-Duhem integration at 773 K. The latter authors determined activities of Cd for the two phase fields: $GdCd + GdCd_2$, $GdCd_2 + GdCd_3$, $GdCd_3 + Gd_{11}Cd_{45}$, $Gd_{11}Cd_{45} + Gd_{13}Cd_{58}$ and $Gd_{13}Cd_{58} + Cd_6Gd$ by isopiestic vapor pressure method.

3 Assessment Procedure

On the basis of Gibbs energies of all pure elements adopted from the study of Dinsdale[12] and the PARROT module[13] of Thermo-Calc software developed by Sundman et al.,[3] the phase relations and the thermochemical properties of the Gd-Cd binary system were carefully assessed based on the experimental data available from literatures.

The PARROT module works by minimizing the square sum of the differences between experimental data and calculated values. In the optimization procedure, each set of experimental data was given a certain weight. The weights were changed systematically during the optimization until most of experimental data was accounted for within the claimed uncertainty limits.

The parameters were optimized step by step. The parameters related to the solution phases, the liquid phase and the terminal solid solution phases, are optimized first. The congruent intermetallic compounds were investigated next. The parameters of the intermetallic compounds are optimized by using phase diagram data and available thermodynamic information. Then experimental data related to invariant equilibria have been added to the calculation. All the parameters were finally evaluated together in order to give the best description of the system.

4 Thermodynamic Models

4.1 Unary Phases

The Gibbs energy of the pure element i (i = Gd, Cd) in the phase ϕ (ϕ = Liquid, hexagonal-close-packed (HCP_A3) and body-centered cubic (BCC_A2), referred to the enthalpy of its stable state at 298.15 K, is described as a function of temperature by:

$$G_i^{\phi} = {}^0G_i^{\phi} - {}^0H_i^{SER}(298.15K) = a + bT + cTlnT + dT^2 + eT^3 + fT^7 + gT^{-1} + hT^{-9} \quad \text{(Eq 1)}$$

where $H_i^{SER}(298.15\ \text{K})$ is the molar enthalpy of the element i at 298.15 K in its standard element reference (SER) state, HCP_A3 for Gadolinium and Cadmium.

In the present work, the Gibbs energy functions are taken from the SGTE (Scientific Group Thermodata Europe) compilation by Dinsdale.[12]

4.2 Solution Phases

The solution phases (liquid, HCP_A3(α-Gd) and BCC_A2 (β-Gd)) were modeled as substitutional solution. The Gibbs energy of one mol of formula unit of phase φ is expressed as the sum of the reference part ${}^{ref}G^{\varphi}$, the ideal part ${}^{id}G^{\varphi}$, and the excess part ${}^{ex}G^{\varphi}$:

$${}^0G^{\varphi} = {}^{ref}G^{\varphi} + {}^{id}G^{\varphi} + {}^{ex}G^{\varphi} \quad \text{(Eq 2)}$$

As used in the Thermo-Calc package:[5]

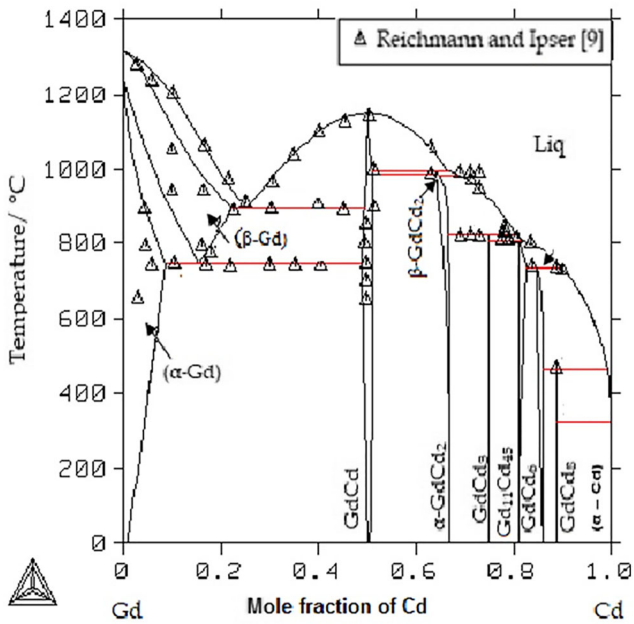

Fig. 4 Gd-Cd phase diagram calculated using the current set of optimized parameters compared with experimental data from Ref 9

$$^{ref}G^{\varphi} = \left[{}^{0}G^{\varphi}_{Gd} - H^{SER}_{Gd}(298.15)\right]x_{Gd} + \left[{}^{0}G^{\varphi}_{Cd} - H^{SER}_{Cd}(298.15)\right]x_{Cd} \quad \text{(Eq 3)}$$

$$^{id}G^{\varphi} = RT[x_{Gd}\ln(x_{Gd}) + x_{Cd}\ln(x_{Cd})] \quad \text{(Eq 4)}$$

where R is the gas constant, T is the temperature, in Kelvin,x_{Gd} and x_{Cd} are the fraction of elements Gd and Cd, respectively.

The excess terms of all the solution phases were modeled by the Redlich-Kister[14] formula:

$$^{ex}G^{\varphi}(T) = x_{Gd}x_{Cd}\sum_{\lambda=0}^{n} {}^{\lambda}L^{\varphi}_{Gd,Cd}(x_{Gd} - x_{Cd})^{\lambda} \quad \text{(Eq 5)}$$

$$^{i}L^{\varphi}_{Gd,Cd} = a_i + b_iT \quad \text{(Eq 6)}$$

where $^{i}L^{\varphi}_{Gd,Cd}(T)$ is the interaction parameter between the elements Ga and Cd, which is evaluated in the present work the coefficients a_i and b_i are optimized using the PARROT module.[13]

4.3 Stoichiometric compounds

The Gibbs energy of the stoichiometric compounds Gd_P-Cd_q ($GdCd_3$ and $GdCd_8$)is expressed as follows:

$$^{0}G_{Gd_pCd_q} = \frac{p}{p+q}{}^{0}G_{Gd} + \frac{q}{p+q}{}^{0}G_{Cd} + a + \text{bT} \quad \text{(Eq 7)}$$

where $^{0}G_{Gd}$ and $^{0}G_{Cd}$ and are the Gibbs energy of the pure Gd and Cd elements respectively. The coefficients aand b which represents respectively the enthalpy and entropies of formation of the Ga_pLi_q compound were evaluated in the present work based on the literature data.

4.4 Intermediate Phases with Homogeneity Range

The intermetallic compounds Gd_mCd_n (β-$GdCd_2$, α-$GdCd_2$ and $GdCd_6$), which have a homogeneity range, were treated by a two-sublattice model as $(Gd)_m(Gd, Cd\%)_n$where $\left(m = \frac{p}{p+q} and n = \frac{q}{p+q}\right)$. In addition, the phase GdCd is a B2 phase, isotypic with CsCl while the phase$Gd_{11}Cd_{45}$- isotypic with $Cd_{45}Sm_{11}$. Both intermetallic compounds have homogeneity ranges, and were treated as (Gd%, $Cd)_m(Gd, Cd\%)_n$ by a two-sublattice model.

The symbol % denotes the major component in the corresponding sublattice.

The Gibbs energy function per mole (m) of the formula unit (Gd%, Cd $)_m(Gd, Cd\%)_n$is the following:

$$\begin{aligned} G^{Gd_mCd_n} - H^{SER}_{Gd_mCd_n} &= y^1_{Gd}y^2_{Cd}{}^{0}G^{Gd:Cd} + y^1_{Gd}y^2_{Gd}{}^{0}G^{Gd:Gd} \\ &+ y^1_{Cd}y^2_{Gd}{}^{0}G^{Cd:Gd} + y^1_{Cd}y^2_{Cd}{}^{0}G^{Cd:Cd} \\ &+ RT\left[m\left(y^1_{Gd}\ln y^1_{Gd} + y^1_{Cd}\ln y^1_{Cd}\right)\right. \\ &\left.+ n\left(y^2_{Gd}\ln y^2_{Gd} + y^2_{Cd}\ln y^2_{Cd}\right)\right] + {}^{exc}G^{Gd_mCd_n}_{m} \end{aligned} \quad \text{(Eq 8)}$$

$$H^{SER}_{Gd_mCd_n} = x_{Gd} * H^{SER}_{Gd} + x_{Cd} * H^{SER}_{Cd} \quad \text{(Eq 9)}$$

where y^1_{Gd}andy^1_{Cd} denote the site fraction of Gd and Cd in the first sublattice, y^2_{Gd} and y^2_{Cd} the site fractions of Gd and Cd in the second sublattice, $^{0}G^{Gd:Gd}$ represents the Gibbs free energy of the hypothetical compound Gd_mGd_n.$^{0}G^{Gd:Cd}$ is the Gibbs energy of the stoichiometric compound Gd_mCd_n.^{exc}G is the excess Gibbs energy given by the following expression:

$$^{exc}G = y^2_{Gd}y^2_{Gd}\left(y^2_{Gd} - y^2_{Cd}\right)^{\lambda}L^{\lambda}_{Gd:Gd,Cd} \quad \text{(Eq 10)}$$

where $L^{\lambda}_{Gd:Gd,Cd}$ (λ= 0, 1, 2...) represents the interaction parameters between the elements Gd and Cd in the second sublattice while the first sublattice is only occupied by the element Gd.These excess parameters can be temperature dependent as for example:

$$L^{\lambda}_{Gd:Gd,Cd} = a_{\lambda} + b_{\lambda}T \quad \text{(Eq 11)}$$

a_{λ} and b_{λ} are the coefficients to be optimized.

For the intermetallic compounds GdCd and $Gd_{11}Cd_{45}$ the Wagner-Schottky model[15] was applied as follows:

$$^{0}G_{Gd_pCd_q} = \frac{p}{p+q}{}^{0}G_{Gd} + \frac{q}{p+q}{}^{0}G_{Cd} + 10000 - a - \text{bT} \quad \text{(Eq 12)}$$

a and b are the coefficients to be optimized.

Table 2 The optimized thermodynamic parameters of the Gd-Cd system

Phase	Thermodynamic models	Parameters in J/mol
Liquid	**(Cd, Gd)$_1$**	${}^0L^{liq} = -101797.713 - 34.831*T$
		${}^1L^{liq} = -54722.463 + 27.805*T$
		${}^2L^{liq} = -15020.506 + 26.561*T$
HCP_A3	**(Cd, Gd)$_1$(Va)$_{0.5}$**	${}^0L^{HCP_A3} = -98130.906 + 6.191 * T$
		${}^1L^{HCP_A3} = 10791.305$
BCC_A2	**(Cd, Gd)$_1$(Va)$_3$**	${}^0L^{BCC_A2} = -99438.4431 - 37.922*T$
		${}^1L^{BCC_A2} = -35633.100$
CdGd	**(Cd, Gd)$_{0.5}$: (Cd, Gd)$_{0.5}$**	$G^{CdGd}_{Gd:Gd} - {}^{298}H^{HCP_A3}_{Gd} = G^{HCP_A3}_{Gd} + 5000$
		$G^{CdGd}_{Cd:Gd} - 0.5{}^{298}H^{HCP_A3}_{Cd} - 0.5{}^{298}H^{HCP_A3}_{Gd}$
		$= 0.5G^{HCP_A3}_{Cd} + 0.5G^{HCP_A3}_{Gd} - 59837.381 + 7.176*T$
		$G^{CdGd}_{Cd:Gd} - 0.5{}^{298}H^{HCP_A3}_{Gd} - 0.5{}^{298}H^{HCP_A3}_{Cd}.$
		$G^{HCP_A3}_{Cd} + 0.5G^{HCP_A3}_{Gd} + 10000 + 59837.381 - 7.176*T$
		$G^{CdGd}_{Cd:Cd} - {}^{298}H^{HCP_A3}_{Cd} = G^{HCP_A3}_{Cd} + 5000$
		${}^0L^{CdGd}_{Gd:Cd,Gd} = 1201.080 + 8.456*T$
		${}^0L^{CdGd}_{Cd,Gd:Cd} = -16019.2$
β-Cd_2Gd	**(Cd, Gd)$_{0.667}$: (Gd)$_{0.333}$**	$G^{\beta-Cd2Gd}_{Gd:Gd} - {}^{298}H^{HCP_A3}_{Gd} = G^{HCP_A3}_{Gd} + 5000$
		$G^{\beta-Cd2Gd}_{Cd:Gd} - 0.667{}^{298}H^{HCP_A3}_{Cd} - 0.333{}^{298}H^{HCP_A3}_{Gd}$
		$= 0.333G^{HCP_A3}_{Gd} + 0.667G^{HCP_A3}_{Cd} - 43882.004 + 0.399*T$
		${}^0L^{\beta-Cd2Gd}_{Gd:Cd,Gd} = -19898.61$
		${}^1L^{\beta-Cd2Gd}_{Gd:Cd,Gd} = -44998.213$
α-Cd_2Gd	**(Gd, Cd)$_{0.667}$: (Gd)$_{0.333}$**	$G^{\alpha-Cd2Gd}_{Gd:Gd} - {}^{298}H^{HCP_A3}_{Gd} = G^{HCP_A3}_{Gd} + 5000$
		$G^{\alpha-Cd2Gd}_{Cd:Gd} - 0.667{}^{298}H^{HCP_A3}_{Cd} - 0.333{}^{298}H^{HCP_A3}_{Gd}$
		$= 0.667G^{HCP_A3}_{Cd} + 0.333G^{HCP_A3}_{Gd} - 43899.975 + 0.414*T$
		${}^0L^{\alpha-Cd2Gd}_{Gd:Cd,Gd} = -19899.954$
		${}^1L^{\alpha-Cd2Gd}_{Gd:Cd,Gd} = -45004.621$
Cd_3Gd	**(Cd)$_{0.75}$:(Gd)$_{0.25}$**	$G^{Cd3Gd}_{Cd:Gd} - 0.75{}^{298}H^{HCP_A3}_{Cd} - 0.25{}^{298}H^{HCP_A3}_{Gd}$
		$= 0.75G^{HCP_A3}_{Cd} + 0.25G^{HCP_A3}_{Gd} - 36213.242 - 0.035*T$
$Cd_{45}Gd_{11}$	**(Cd, Gd)$_{0.81}$:(Cd, Gd)$_{0.19}$**	$G^{Cd45Gd11}_{Gd:Gd} - {}^{298}H^{HCP_A3}_{Gd} = G^{HCP_A3}_{Gd} + 5000$
		$G^{Cd45Gd11}_{Cd:Gd} - 0.81{}^{298}H^{HCP_A3}_{Cd} - 0.19{}^{298}H^{HCP_A3}_{Gd}$
		$= 0.81G^{HCP_A3}_{Cd} + 0.19G^{HCP_A3}_{Gd} - 30414.500 - 0.141*T$
		$G^{Cd45Gd11}_{Gd:Cd} - 0.19{}^{298}H^{HCP_A3}_{Cd} - 0.81{}^{298}H^{HCP_A3}_{Gd}$
		$= 0.19G^{HCP_A3}_{Cd} + 0.81G^{HCP_A3}_{Gd} + 10000 + 30414.500 + 0.141*T$
		$G^{Cd45Gd11}_{Cd:Cd} - {}^{298}H^{HCP_A3}_{Cd} = G^{HCP_A3}_{Cd} + 5000$
		${}^0L^{Cd45Gd11}_{Gd:Cd,Gd} = 622.255$
		${}^0L^{Cd45Gd11}_{Cd,Gd:Cd} = -10015.214$
Cd_6Gd	**(Cd, Gd)$_{0.86}$: (Gd)$_{0.14}$**	$G^{Cd6Gd}_{Gd:Gd} - {}^{298}H^{HCP_A3}_{Gd} = G^{HCP_A3}_{Gd} + 5000$
		$G^{Cd6Gd}_{Cd:Gd} - 0.86{}^{298}H^{HCP_A3}_{Cd} - 0.14{}^{298}H^{HCP_A3}_{Gd}$
		$= 0.86G^{HCP_A3}_{Cd} + 0.14G^{HCP_A3}_{Gd} - 24734.763 + 0.338*T$
		${}^0L^{Cd6Gd}_{Gd:Cd,Gd} = -55052.109$
		${}^1L^{Cd6Gd}_{Gd:Cd,Gd} = -58955.606$
Cd_8Gd	**(Cd)$_{0.889}$:(Gd)$_{0.111}$**	$G^{Cd8Gd}_{Cd:Gd} - 0.889{}^{298}H^{HCP_A3}_{Cd} - 0.111{}^{298}H^{HCP_A3}_{Gd}$
		$= 0.889G^{HCP_A3}_{Cd} + 0.111G^{HCP_A3}_{Gd} - 19568.483 - 0.274*T$

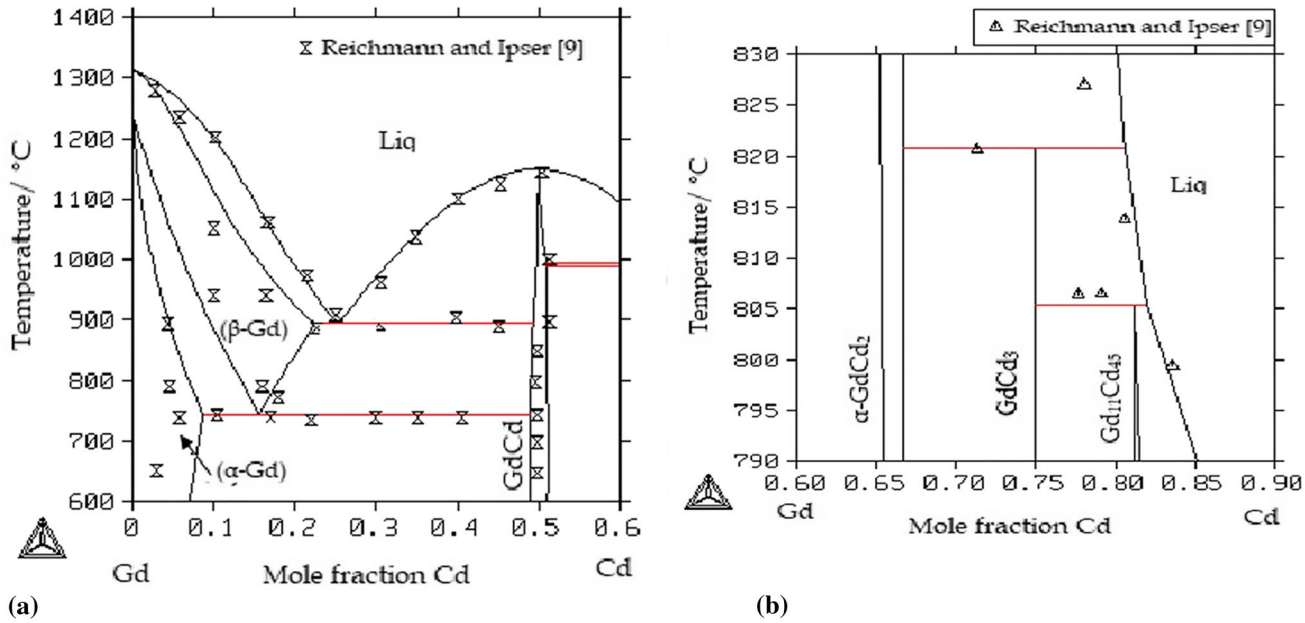

Fig. 5 (a) Gd-rich region of the Gd-Cd phase diagram calculated using the current set of optimized parameters compared with experimental data from Ref 9. (b) Part of the Gd–Cd phase diagram showing the peritectic reaction L + $GdCd_3 \leftrightarrow Gd_{11}Cd_{45}$

5 Results and Discussion

The thermodynamic optimization of the Gd-Cd system is carried out in the present work using the PARROT module in the Thermo–Calc software,[3] taking into account all the relevant experimental results published by Reichmann and Ipser[9] and the enthalpies of formation of all intermetallic compounds measured by Roshchina and Bayanov[10] and calculated by Kurata and Sakamura.[7] Figure 4 shows the calculated equilibrium phase diagram with the numerous experimental measured by Reichmann and Ipser[9] used in this optimization.

The calculated phase diagram agrees well with most of the experimental data. We noted that our calculated values concerning invariant reactions of four intermetallic compounds, β-$GdCd_2$, $GdCd_3$, $Gd_{11}Cd_{45}$ and $GdCd_6$, which are formed by peritectic reactions in this system, are in a good agreement with the temperatures and compositions determined experimentally by Reichmann and Ipser.[9]

For the GdCd phase, composition data are inconclusive (~ 49.8 to ~ 50.9. at.% Cd). The calculated congruent-melting point of this phase is 1153 °C which is in a good agreement with the work of Reichmann and Ipser.[9] The liquidus calculated in general is in the best agreement with the experimental data measured by Reichmann and Ipser.[9]

The optimized thermodynamic parameters for all then phases in the Gadolinium-Cadmium binary system obtained are given in Table 2.

The Gd-rich region of the Gd-Cd phase diagram and detail of the eutectoid reactions are plotted in Fig. 5(a). In Fig. 5(b) the magnification shows the peritectic reaction L + $GdCd_3 \leftrightarrow Gd_{11}Cd_{45}$.

The invariant reactions in the Gd-Cd system are listed in Table3 and compared with the results by Reichmann and Ipser[9] and Gschneidner and Calderwood.[5] They are in satisfactory agreement.

As mentioned previously, the enthalpies of formation of intermetallic compounds are measured by Roshchina and Bayanov[10] by electromotive force method and calculated by Kurata and Sakamura,[7] so these values were used in the present study with high weights. The assessed enthalpies of formation of Gd-Cd intermetallic compounds compared with experimental measurements are listed in Table 4, also presented in Fig. 6. A very good agreement was obtained.

The values of the Gibbs energies of formation of the phases $GdCd_6$ $Cd_{58}Gd_{13}$, $Cd_{45}Gd_{11}$, Cd_3Gd and Cd_2Gd have been placed in Fig. 7. Received values are in a satisfactory agreement with previous data obtained by Reichmann et al[11] and Roshchina and Bayanov.[10]

The evolution of the activities of Cd at 773 K calculated is shown in Fig. 8. and compared values determined experimentally by Reichmann et al.[11] A good agreement is noted.

As first stated by Kaptay,[16] to check whether the optimized thermodynamic parameters of the intermetallic compounds are satisfactory, we verified that when the

Table 3 Calculated invariant reactions compared with the literature data in the Gd–Cd system

Reaction	Reaction type	Literature data[5, 9]					This work			
		Phase compositions, at.% Cd			T, °C	Reference	Phase compositions, at.% Cd			T, °C
(βGd) ↔ CdGd + (αGd)	Eutectoid	16	50	…	725	[5]	15.5	48.9	8.5	744.8
		19.1	49.6	4.6	745 ± 4	[9]				
L ↔ CdGd + (βGd)	Eutectic	27	50	22	937	[5]	26.9	49.3	22.3	892.9
		26.5	49	22.6	895 ± 5	[9]				
CdGd ↔ L	Congruent	…	50	…	1170	[5]	…	50	…	1150
	melting	…	50	…	1147 ± 5	[9]				
α- Cd_2Gd ↔ β- Cd_2Gd	Polymorphic	…			995	[5]	…			…
	transformation				986 ± 3	[9]				
β- Cd_2Gd + CdGd ↔ α- Cd_2Gd	Peritectoid						64.2	51.3	64.2	989.1
β- Cd_2Gd ↔ L + α- Cd_2Gd	Metatectic						64.5	68.1	64.9	984.1
L + CdGd ↔ β- Cd_2Gd	Peritectic	67.2	50	66.7	1010	[5]	65.9	51.2	64.2	994.8
		68.3	51.2	66.1	998 ± 4	[9]				
L + α- Cd_2Gd ↔ Cd_3Gd	Peritectic	79	66.7	75	815	[5]	78.9	66.6	75	829.8
		79	67.1	75	819 ± 3	[9]				
L + Cd_3Gd ↔ $Cd_{45}Gd_{11}$	Peritectic	81.6	75	80.4	806	[5]	82.4	75	81	782.6
		81.5	75	80.2	808 ± 2	[9]				
L + $Cd_{45}Gd_{11}$ ↔ $Cd_{58}Gd_{13}$	Peritectic	82.4	80.4	81.7	803	[5]	…			
L + $Cd_{58}Gd_{13}$ ↔ Cd_6Gd	Peritectic	88.6	81.7	85.7	716	[5]				
L + $Cd_{45}Gd_{11}$ ↔ Cd_6Gd	Peritectic	88.2	81.7	85.7	730 ± 2	[9]	89.5	82.7	84.5	694
L ↔ (Cd) + Cd_6Gd	Eutectic	97.5	100	85.7	316	[5]	…			
L + Cd_6Gd ↔ Cd_8Gd	Peritectic	99	85.7	88.9	465 ± 2	[9]	94.4	85.6	88.9	579.5
L ↔ (Cd) + Cd_8Gd	Eutectic	99.8	100	88.9	320 ± 2	[9]	99.6	100	88.9	319

Table 4 Calculated and measured enthalpies of formation of the intermetallic compounds

Intermetallic compound	Enthalpy of formation, kJ/mol. at	T/K	Technique used	Reference
CdGd	− 55.65	298	Optimization	[7]
	− 60.83	298	Optimization	This work
β-Cd_2Gd	− 43.88	298	Optimization	This work
α-Cd_2Gd	− 42.96	298	Optimization	[7]
	− 43.89	298	Optimization	This work
Cd_3Gd	− 37.17	298	Optimization	[7]
	− 36.07	298	Optimization	This work
$Cd_{45}Gd_{11}$	− 33.10	298	Optimization	[7]
	− 30.41	298	Optimization	This work
$Cd_{58}Gd_{13}$	− 31.92	298	Optimization	[7]
Cd_6Gd	− 28.2	728	Electromotive Force (E.M.F)	[10]
	− 22.9	298		
	− 26.78	298	Optimization	[7]
	− 24.73	298	Optimization	This work
Cd_8Gd	− 19.56	298	Optimization	This work

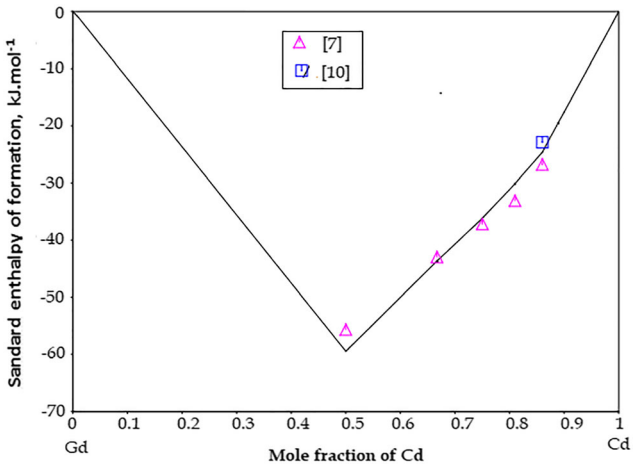

Fig. 6 Calculated standard enthalpies of formation of the intermetallic compounds in the Gd-Cd system compared with the experimental data[7,10]

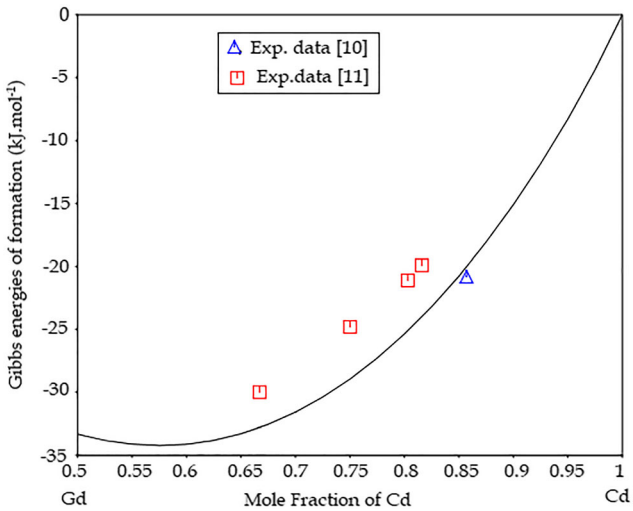

Fig. 7 Calculated the Gibbs energies of formation of the intermetallic compounds in the Gd-Cd system at 773 K compared with the experimental data.[10,11] The reference states are Gd (HCP_A3) and Cd (l)

liquid phase is suspended, the intermetallic compounds are no more stable at high temperatures and disappear at lower temperatures, see Fig. 9. A terminal solid solution and a two phase domain existing between them are found to be stable.

Finally, we verified that no miscibility gap was calculated with our thermodynamic optimized parameters as commended by Chen et al.[17] Figure 10 shows the evolutions of Gibbs energy for the liquid phase as a function of temperature (T), when T is increasing up to 5000 K, the Gibbs energy for the liquid phase decreases. No anomaly is detected on all the curves.

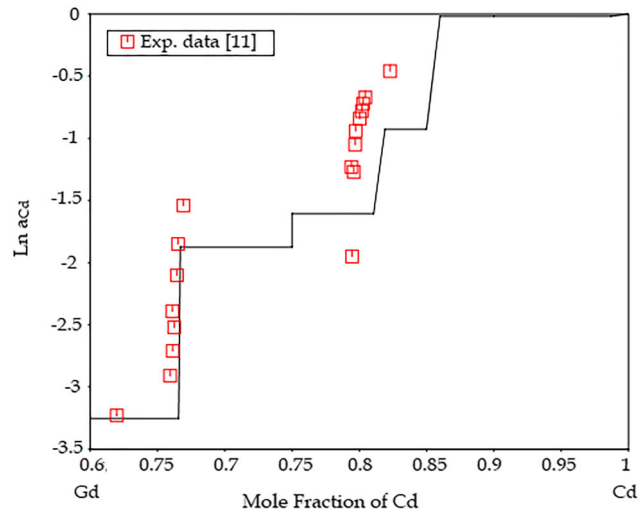

Fig. 8 Comparison of calculated Cd activities with values from literature[11] at 773K. The reference states are Gd (HCP_A3) and Cd (l)

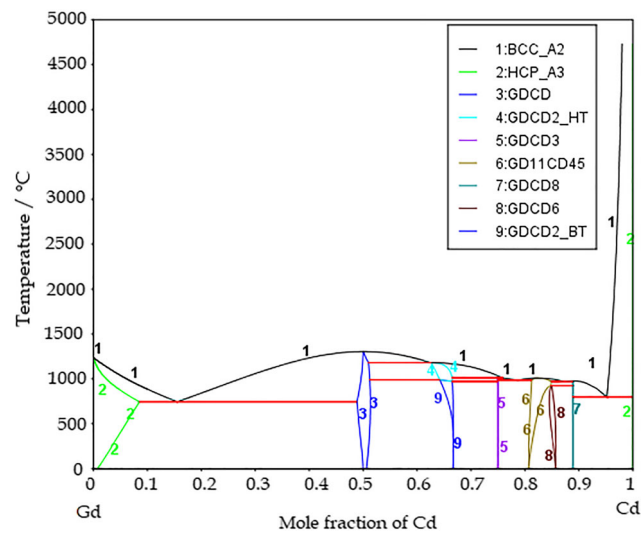

Fig. 9 Calculated Gd–Cd phase diagram when the liquid phase is suspended

6 Conclusion

Based on the critical review of the experimental information on phase diagram and thermodynamic properties in the published literature, thermodynamic assessment of the Gd-Cd binary system has been performed using the CALPHAD method. The calculated phase equilibria agree with the literature data. However, this description should be viewed as a preliminary one. More experimental work on this system may be necessary to improve the description.

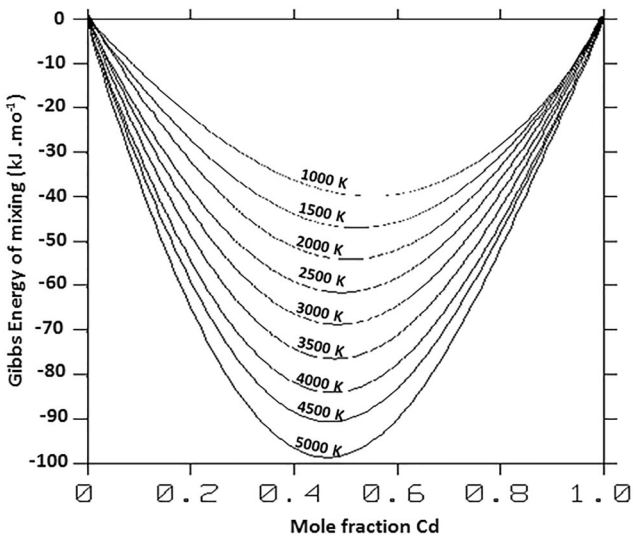

Fig. 10 Gibbs energy of mixing of the liquid phase at different temperature. The reference states were Gd and Cd liquids

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